Selective and Ramsey for $\mathcal{R}_1$ ultrafilters and their Dedekind Cuts

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Outline

1. Topological Ramsey Spaces
   - Definition of a topological Ramsey space
   - The topological Ramsey space $\mathcal{R}_1$.
2. Ultrafilters on $[\mathbb{T}]$
   - Selective and Ramsey for $\mathcal{R}_1$
3. Dedekind Cuts
   - Cuts induced by ultrafilter mappings
   - Dedekind cuts associated to P-point and weakly Ramsey ultrafilters
   - Cuts associated to selective and Ramsey for $\mathcal{R}_1$ ultrafilters

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Example (The Ellentuck Space, $([\omega]^\omega, \subseteq, r)$)

$$r_n(\{a_0, a_1, a_2, \dots\}) = \{a_0, \dots, a_{n-1}\}$$

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A subset $\mathcal{X}$ of $\mathcal{R}$ is **Ramsey** if for every nonempty $[s, X]$, there is a $Y \in [s, X]$ such that $[s, Y] \subseteq \mathcal{X}$ or $[s, Y] \cap \mathcal{X} = \emptyset$.

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The Ellentuck Theorem (E. Ellentuck, 1974 [3])

The Ellentuck space $([\omega]^\omega, \subseteq, r)$ is a topological Ramsey space.

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For each $n < \omega$, let

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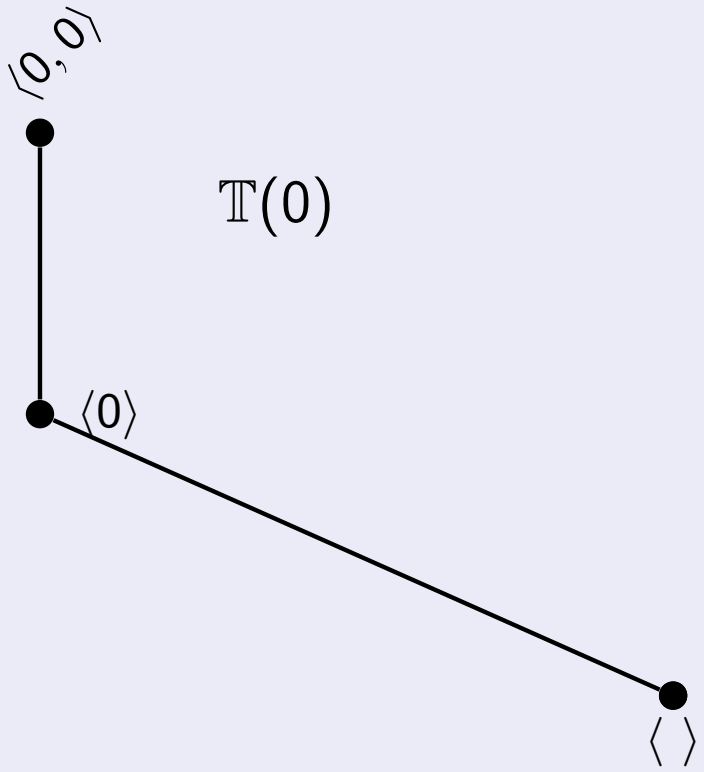

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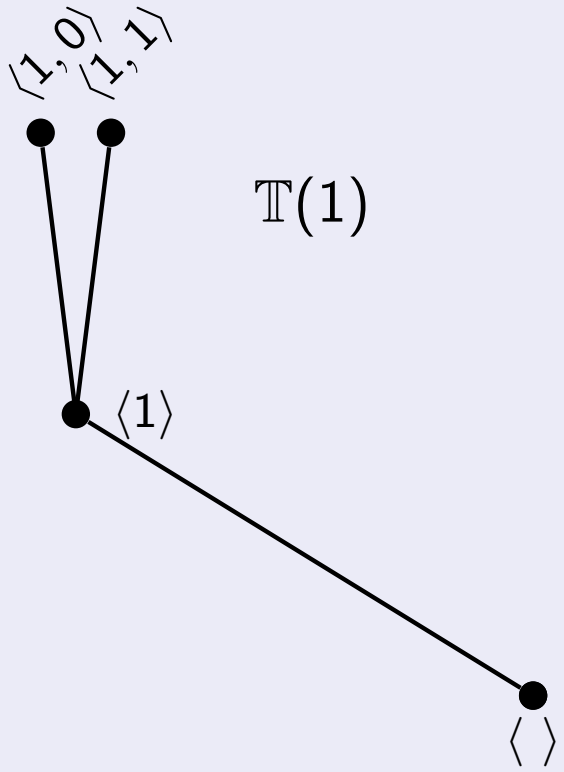

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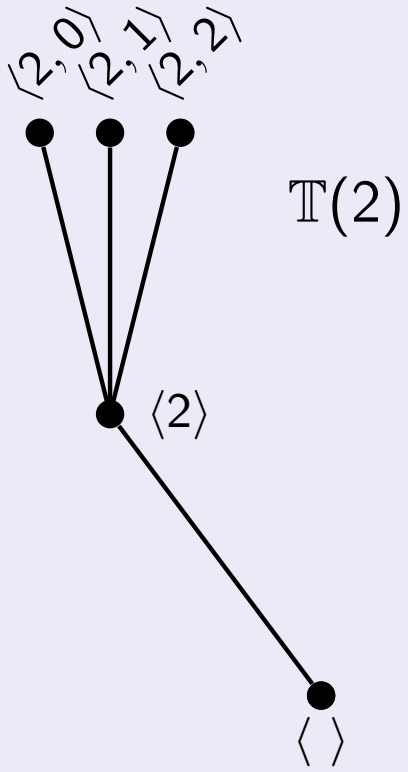

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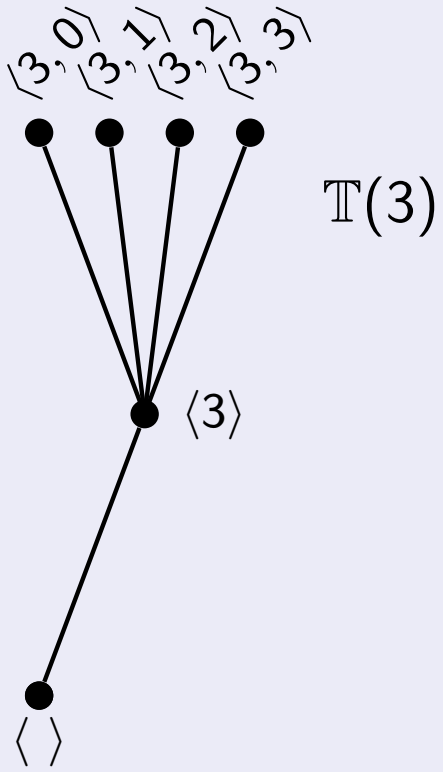

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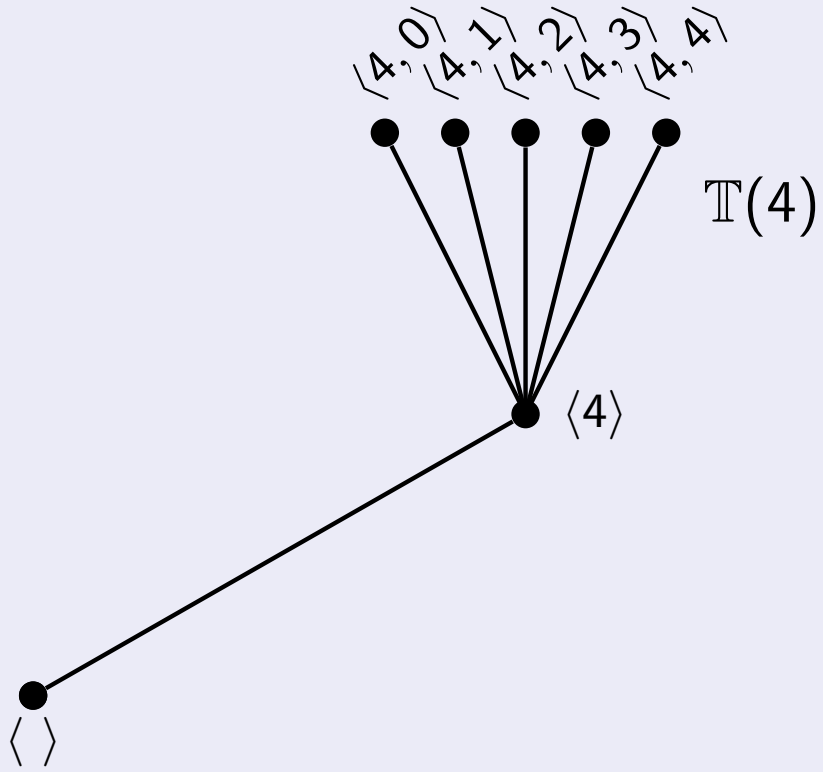

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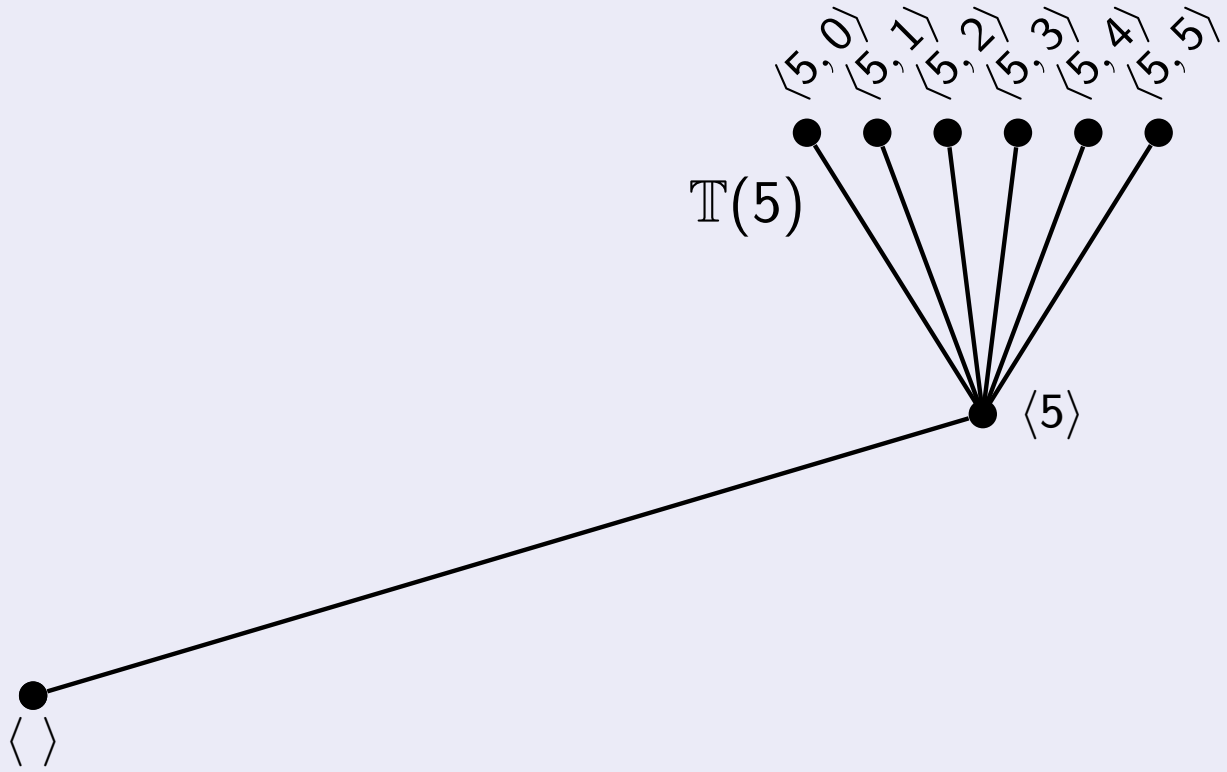

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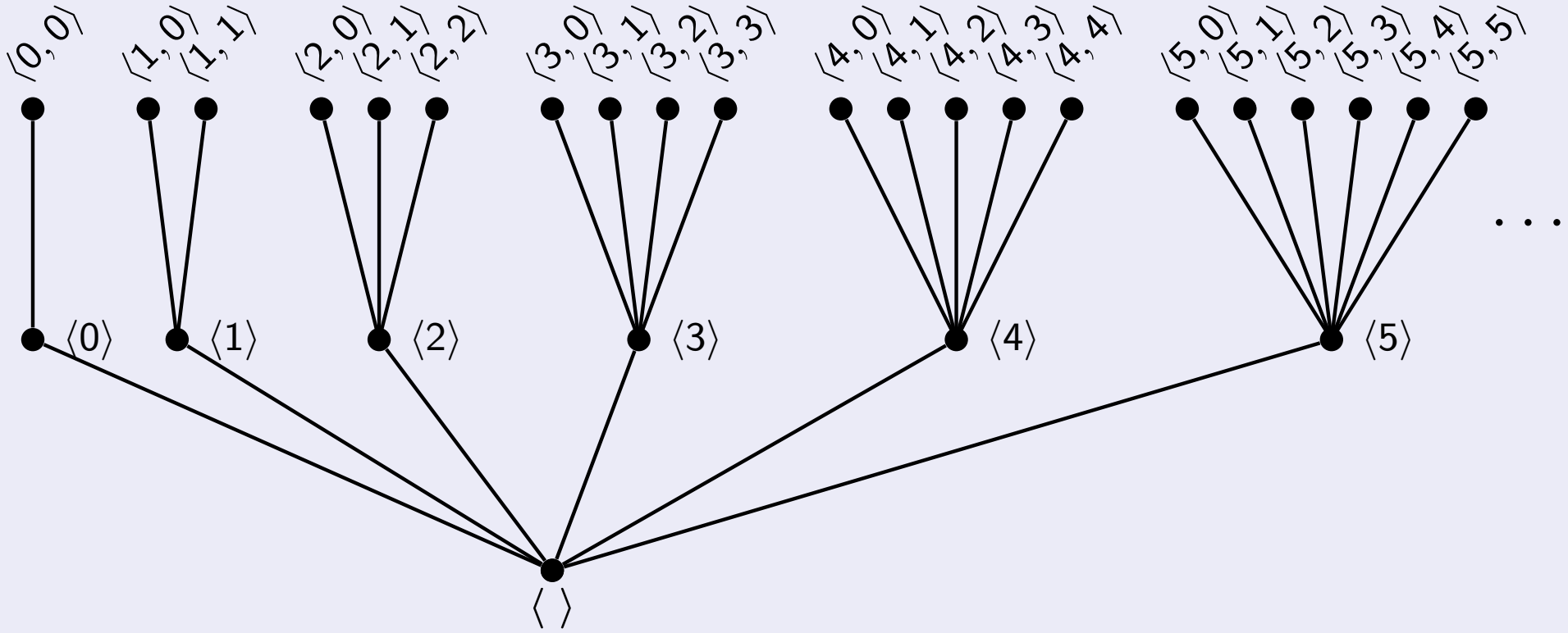

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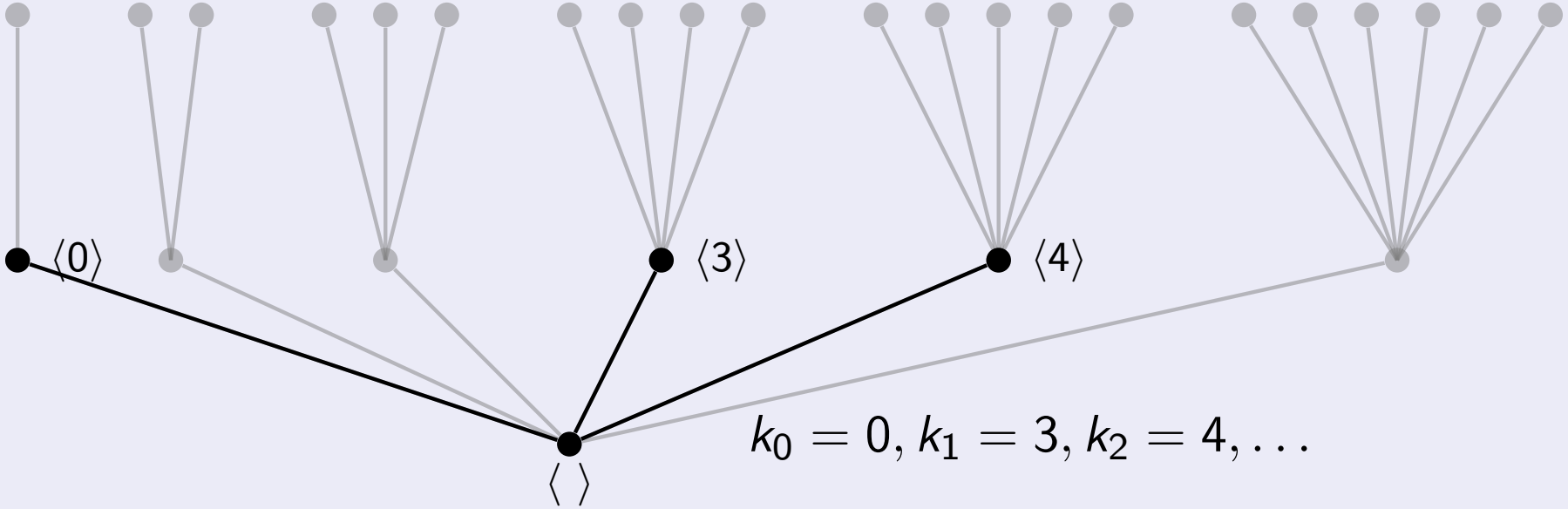

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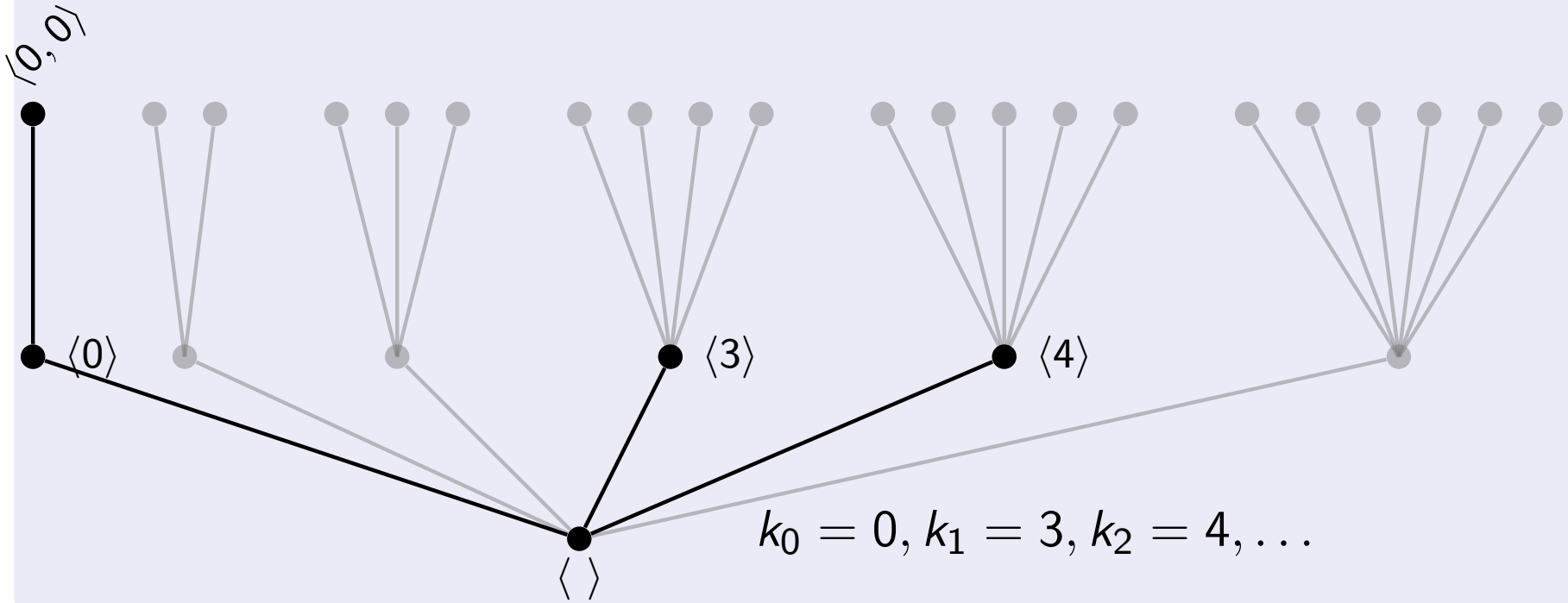

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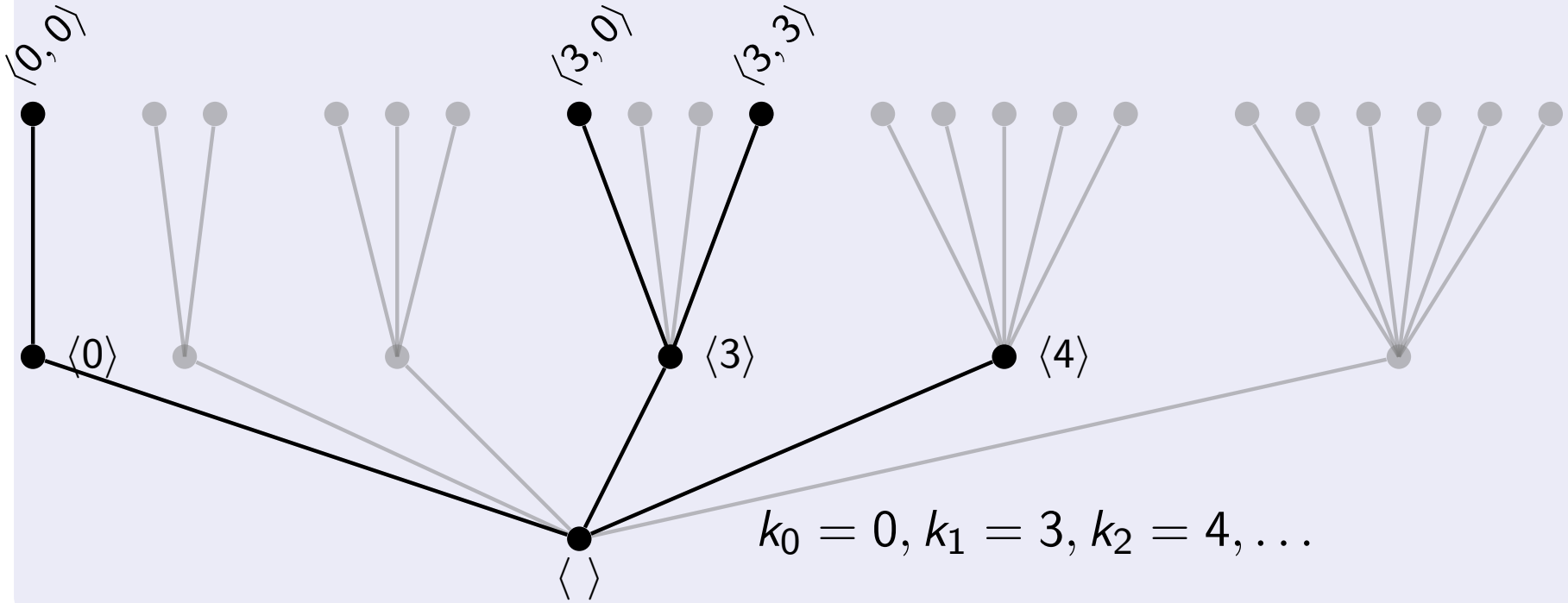

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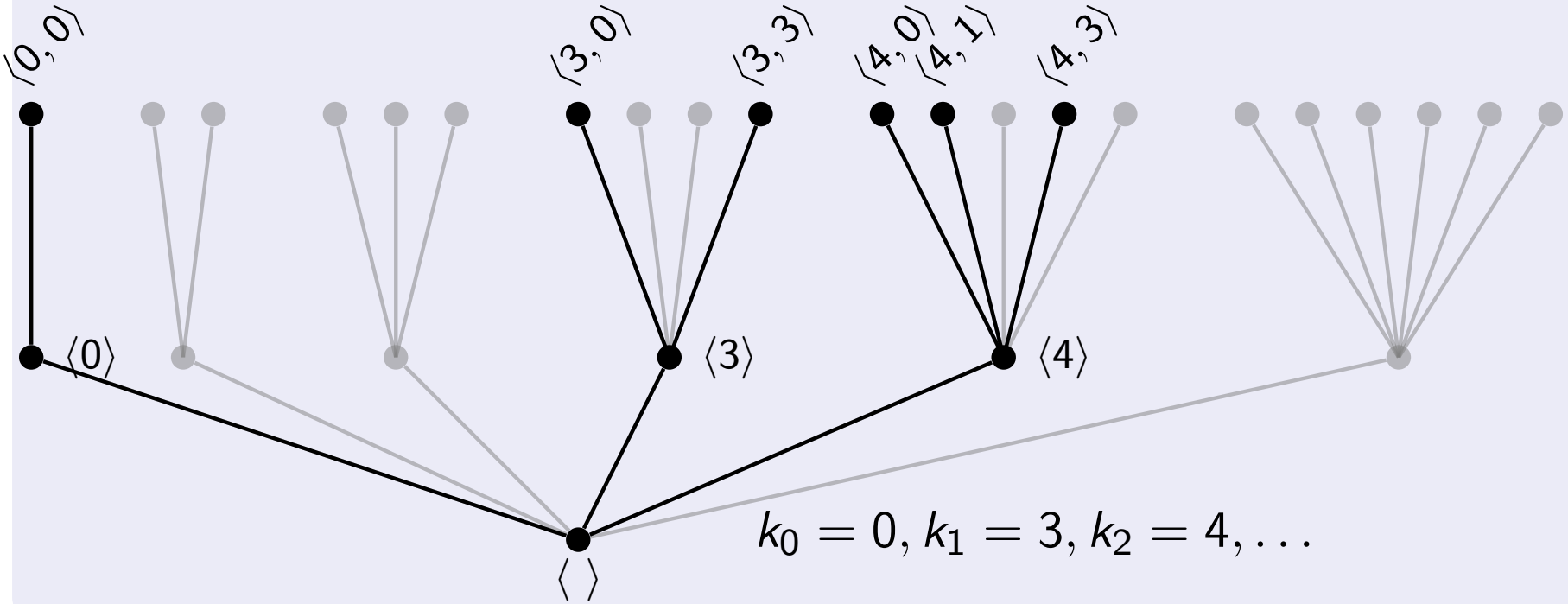

$k_0 = 0, k_1 = 3, k_2 = 4, \ldots$

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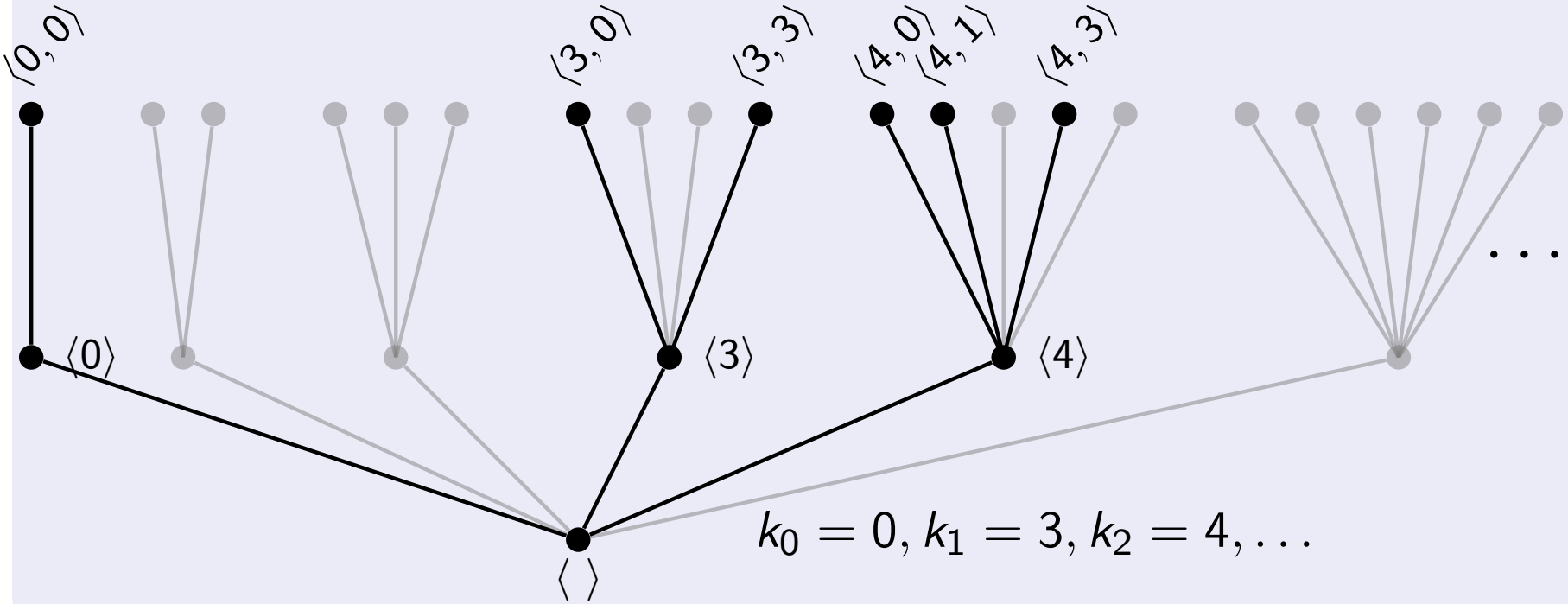

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Theorem (N. Dobrinen and S. Todorcevic, 2013 [2])

$(\mathcal{R}_1, \subseteq, r)$ *is a topological Ramsey space.*

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4. Let $i < j < \omega$. Define $\pi_{T(i)} : \mathcal{R}_1(j) \to \mathcal{R}_1(i)$ to be the map that removes the last $j - i$ branches of a given element of $\mathcal{R}_1(j)$.

Definition (J. Mijares, 2007 [4]; N. Dobrinen, S. Todorcevic, 2013 [2],T., 2013 [6])

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The ultrafilter generated by $\mathcal{C}|R_1(m)$ will be denoted by $\mathcal{U}|\mathcal{R}_1(m)$.

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The set of all equivalence classes of cardinality functions of sets in $\mathcal{U}$, and all larger elements of $\omega^Y/p(\mathcal{U})$, constitute the upper part L of a cut (S, L) of $\omega^Y/p(\mathcal{U})$, which we call **the cut associated to p and $\mathcal{U}$**.

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Assume CH. (S, L) is the cut associated to some map of some P-point to $\mathcal{U}$ if and only if $\mathcal{U}$ is a P-point, S is closed under addition, and every countable subset of L has a lower bound in L.

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Theorem (A. Blass, 1974 [1])

Assume CH. (S, L) is the cut associated to some map of some weakly Ramsey ultrafilter to $\mathcal{U}$ if and only if $\mathcal{U}$ is Ramsey, S is closed under exponentiation, and every countable subset of L has a lower bound in L.

Lemma (T., 2013 [6])

If $\mathcal{U}$ is a selective for $\mathcal{R}_1$ ultrafilter then the cut associated to $\mathcal{U}$ and π is the standard cut in $\omega^\omega/\pi(\mathcal{U})$.

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Let $i < j < \omega$. If $\mathcal{U}$ is a Ramsey for $\mathcal{R}_1$ ultrafilter then the cut associated to $\mathcal{U}|\mathcal{R}_1(j)$ and $\pi_{T(i)}$ is the standard cut in $\omega^{\mathcal{R}_1(i)}/\mathcal{U}|\mathcal{R}_1(i)$.

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Lemma (T., 2013 [6])

Assume CH. If $\mathcal{V}$ is selective then there exists a Ramsey for $\mathcal{R}_1$ ultrafilter $\mathcal{U}$ such that $\pi(\mathcal{U}) = \mathcal{V}$.

Dedekind cuts associated to selective and Ramsey for $\mathcal{R}_1$ ultrafilters

Theorem (T., 2013 [6])

Assume CH. (S, L) is the cut associated to π and some selective for $\mathcal{R}_1$ ultrafilter to $\mathcal{V}$ if and only if $\mathcal{V}$ is selective and (S, L) is the standard cut in $\omega^\omega/\mathcal{V}$.

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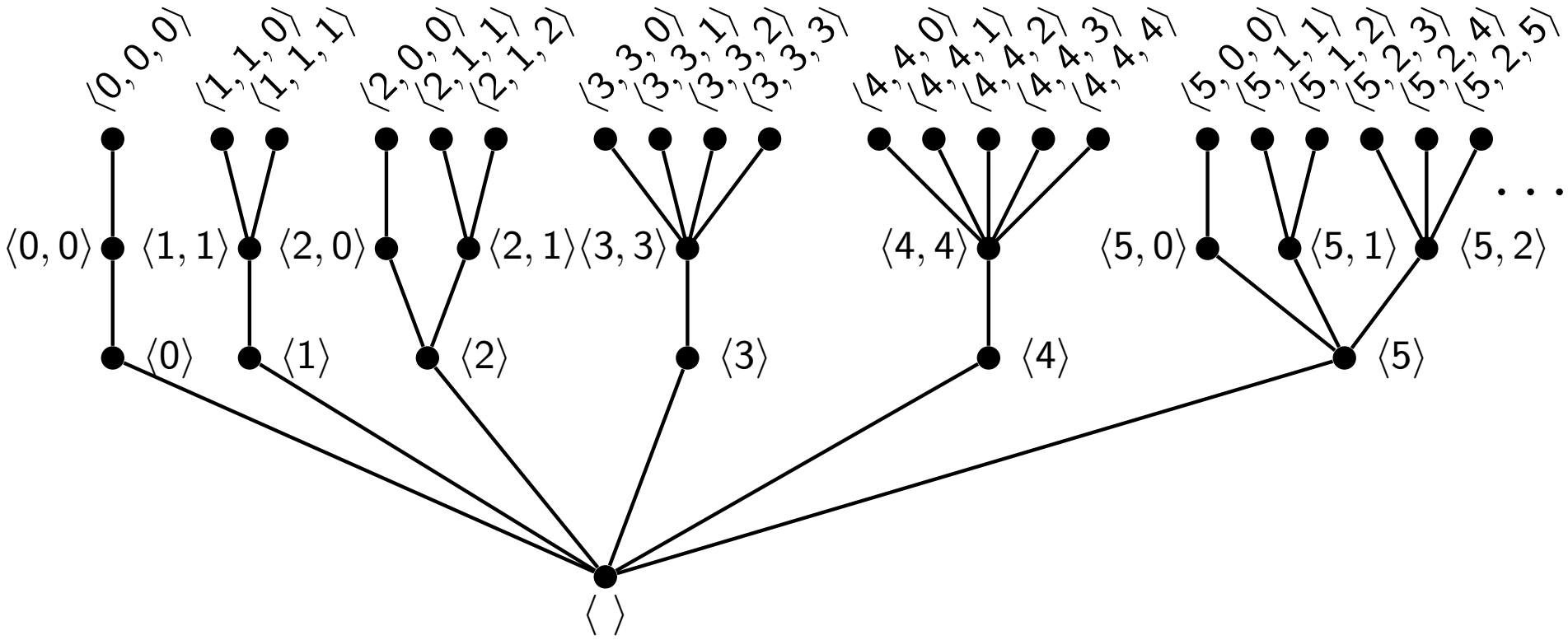
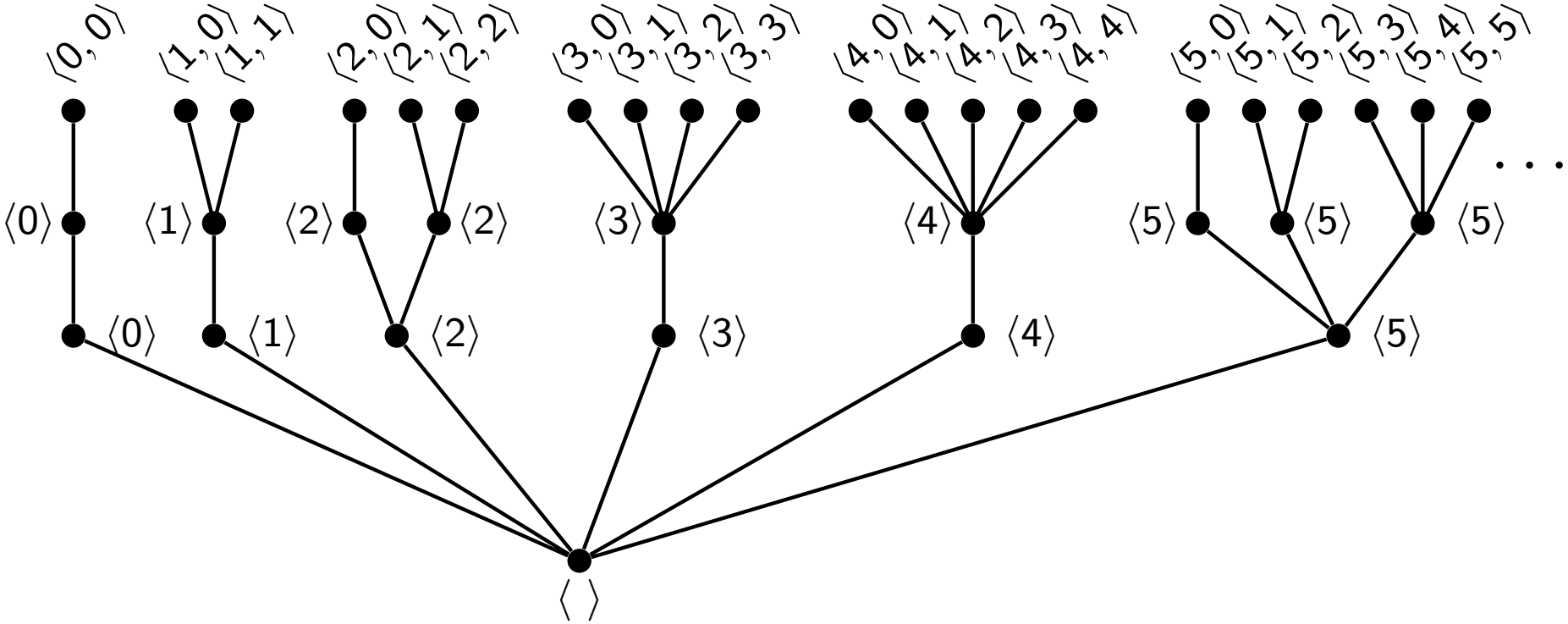


$$F : \mathcal{R}_1(2) \to \{ \text{green}, \text{red} \}$$



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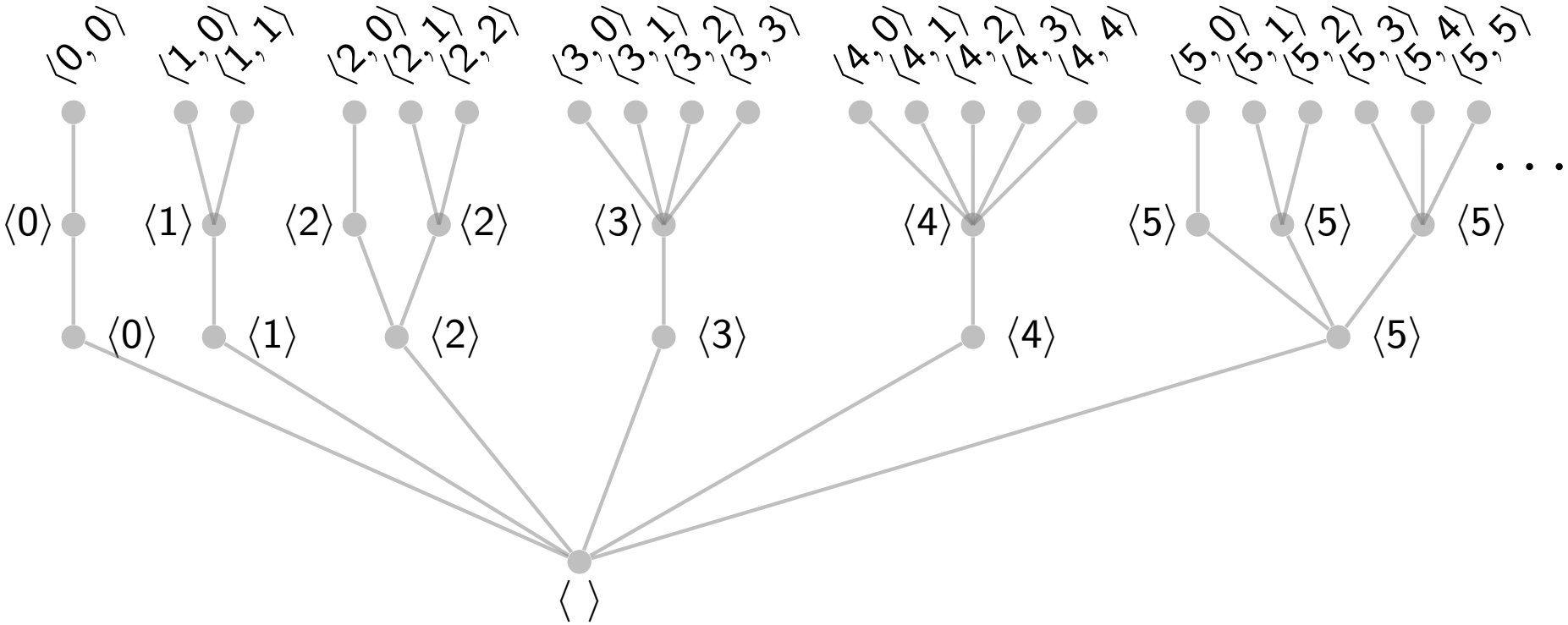

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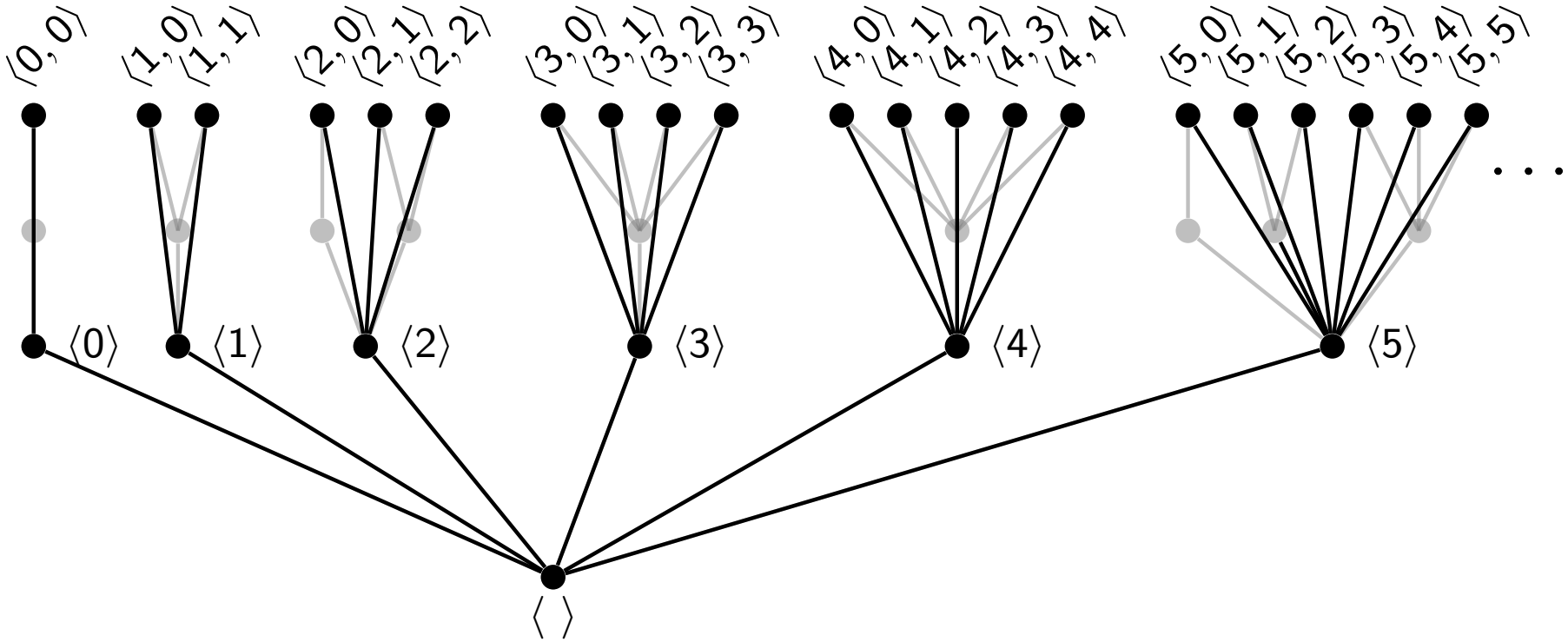

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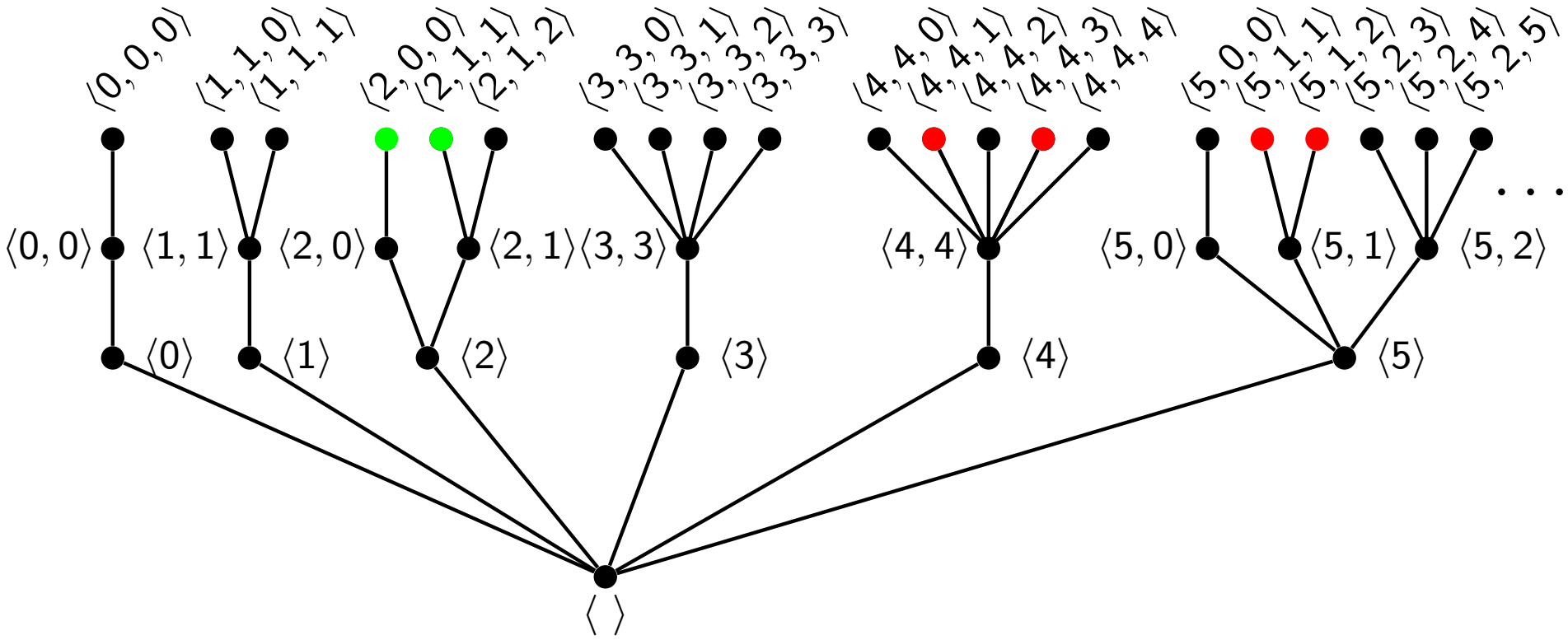

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